Delivery Performance and Customer Satisfaction

Yash Arabhavi · Data Visualization Design Project · Brazilian e-commerce marketplace, 95,824 delivered and reviewed orders

My two questions:

1. How does delivery delay affect customer review scores?
2. Is it being slow, or breaking the promise, that angers customers?

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1. Data and scope

Our cleaned table is `master_orders.csv`, built by `preprocess.py`. One row per order, with the delivery dates, the review score and the customer and seller details already joined. After keeping delivered orders that have a review, I have 95,824 orders to work with.

Where an order has more than one review the pipeline keeps one score for it. I tried the different ways of picking that score and they all land within 0.1 pp of each other, so nothing below depends on it.

Both my questions need a delay, and an order that never showed up does not have one. So both are about delivered orders only. That leaves some very unhappy customers out, and Figure 2 is about them.

I use the **negative rate** (share giving 1 or 2 stars) instead of the average score. The scores are lopsided: 1-star on its own is bigger than 2-star and 3-star put together, so an average of 4.09 does not describe many people.

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2. Question 1: How does delivery delay affect review scores?

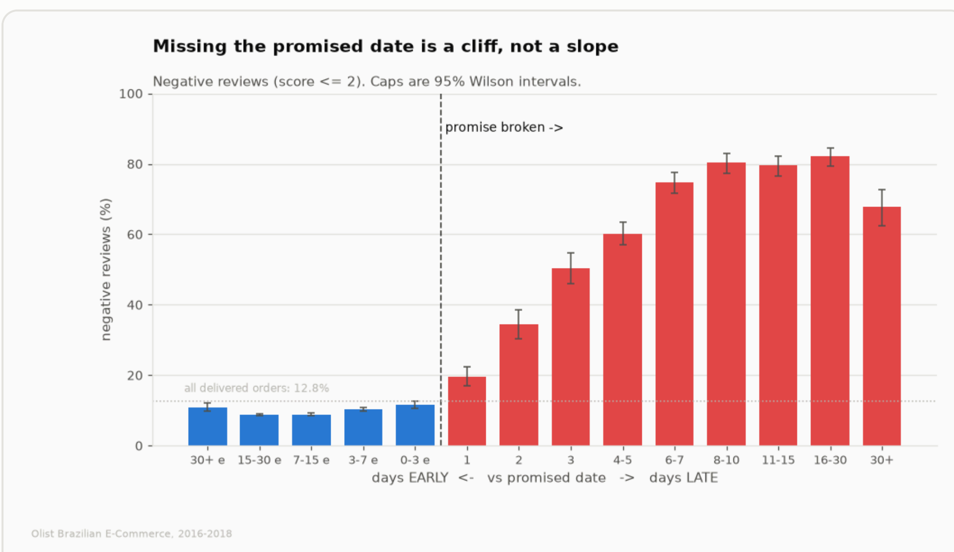

Figure 1. Bad reviews by how early or late the order arrived.

Across sixty days of being *early*, the rate only moves between 8.9% and 11.6%. Arriving three weeks early is barely better than arriving a day early. One day late is 19.6%. Three days late is 50.4%.

2,849 reviewed orders never turned up at all, so Q1 cannot include them. They are 78% negative, against 9.2% for on-time orders.

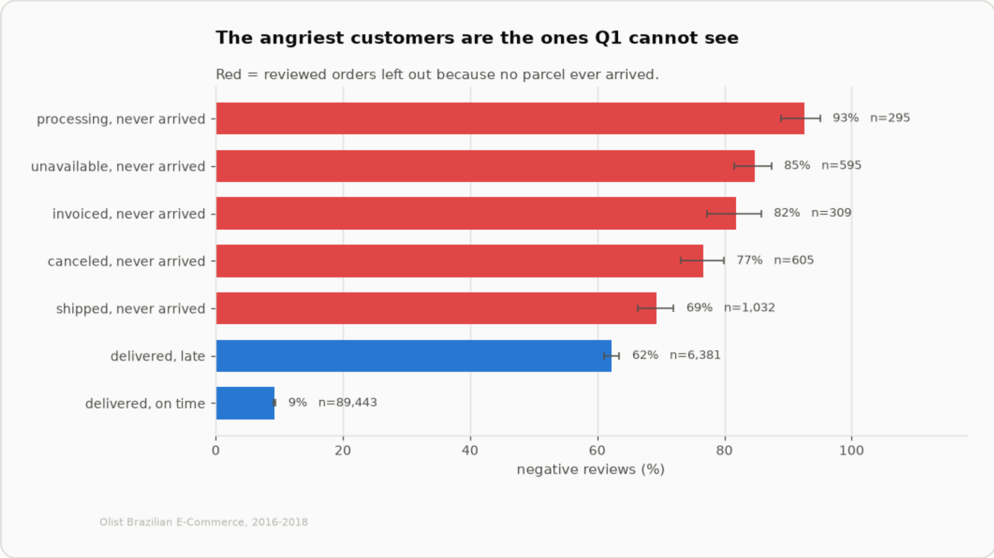

Figure 2. The reviewed orders Q1 has to leave out, next to the two it keeps.

> **Answer to Q1.** Delay does not slowly make people unhappy. It breaks at one point, the promised date. On-time orders are 9.2% negative and late ones are 62.3%, which is 6.8x worse. Late deliveries are only 6.7% of orders but cause 32.5% of all bad reviews. One thing to be fair about: this only covers orders that arrived. The worst case, an order that never comes, is left out before the question starts.

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3. Question 2: Slow, or broken promise?

The obvious test is a 2x2, fast or slow against on time or late. I built it but it does not work. Late orders are nearly always slow as well, so the box that would separate the two only holds 187 orders out of 95,824, with a 95% range of 18.5 to 30.7%. It is in `outputs/q2_two_by_two.csv` but I do not use it for the answer.

Instead I measured the same thing twice on one scale: what an extra day of waiting costs when the order is on time, and what it costs when it is already late.

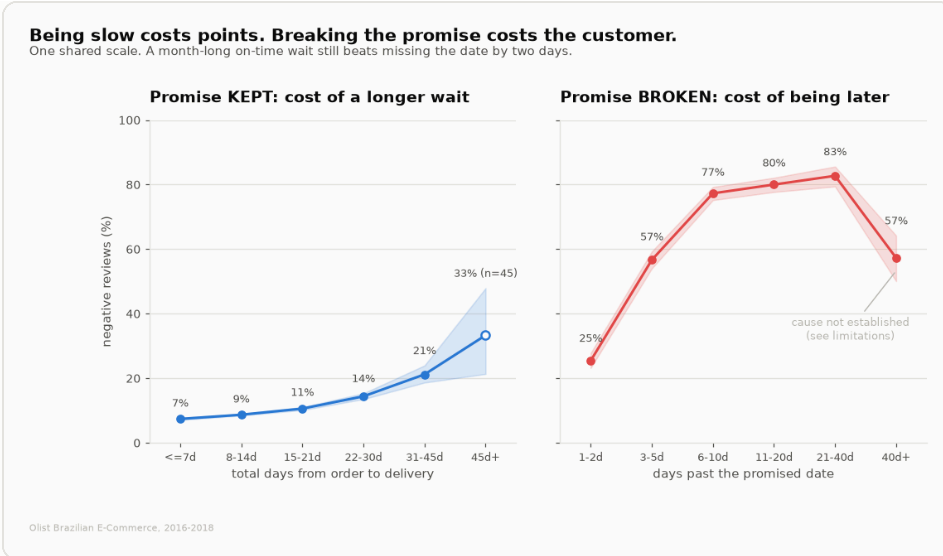

Figure 3. What waiting costs when the order is on time, and when it is late.

When the promise holds, going from under a week to 31-45d moves the rate from 7.4% to 21.2%. When it is broken, the same kind of axis runs from 25.4% to 82.7%. That is 4.2x the range.

While trying to work out why the last late bucket looked odd, I found the reason the two are so different. The review survey is sent when the order is delivered, or on the promised date if it still has not arrived, whichever happens first. Only 0.3% of on-time customers get the survey before their order arrives. For late customers it is 74.5%.

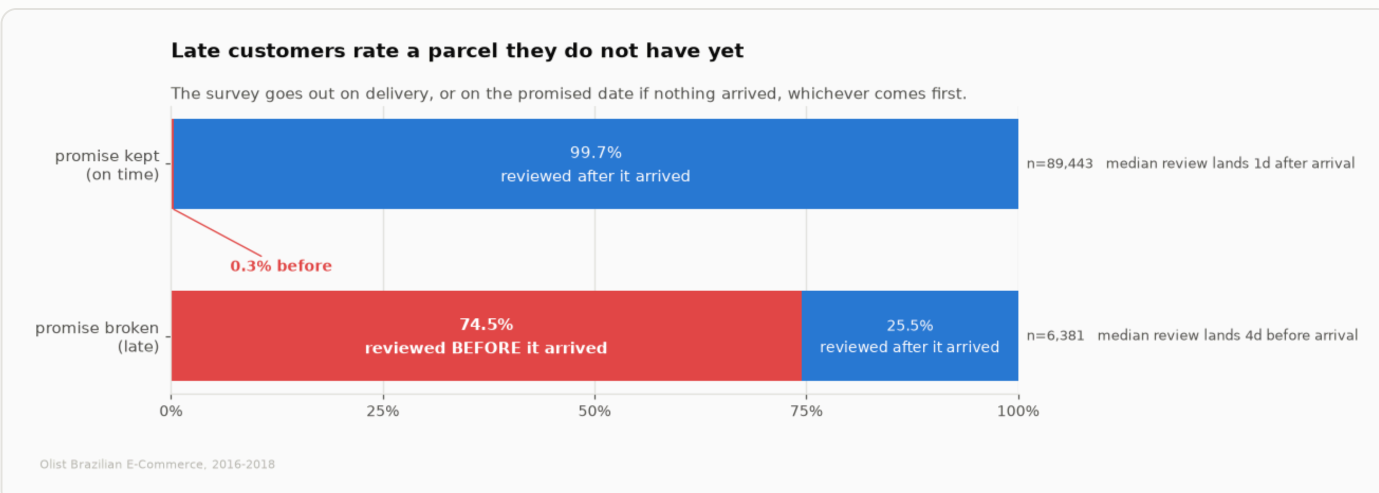

Figure 4. How many of each group were asked before their order arrived.

> **Answer to Q2.** Breaking the promise, clearly. An order that took over a month but arrived on the promised day (21.2% negative) is still better than one that was 1-2d late (25.4%). People are not really rating how long they waited. They are rating the difference between what they were told and what happened. And three out of four late reviews are not even about a late delivery, they are about an order that had not shown up yet.

4. Chart choices

A short note on why each chart looks the way it does.

Figure	Chart	Why
1	bars, split at zero	Bars start from the same line, so lengths are easy to compare. The dashed line at day zero shows the split before you read anything.
2	sideways bars, sorted	Sideways because the labels are long sentences. Sorted worst first, otherwise the reader has to do the sorting.
3	two line charts, same scale	Lines because the x axis is a length of time. Same y scale on both, otherwise each side fills its own box and a 14-point rise looks like a 57-point one.
4	stacked bar of shares	The answer is a share, so I drew a share. I tried a histogram first and it did not work: one bar held 93% of the on-time group and everything else went flat. Each block is labelled with its own number, and the 0.3% sliver is labelled from outside with a line pointing at it, so no number ever sits on a block it does not belong to.

Rules I stuck to

- Only two colours, and each one means something. Blue is the normal group, red is always the bad outcome. I never used red for anything else, so once you have seen one chart you know what red means in the next. Blue and red also stay apart for colour-blind readers, which red and green would not.
- No second y axis. Figure 3 is two charts sharing one scale instead. With separate scales you can make almost any two lines look the same.
- Every rate has an error bar. Some buckets are small, and without the bars a rate built on 45 orders looks as solid as one built on 40,000. Buckets under 100 orders are drawn hollow as an extra warning.
- Labels written on the bars instead of a legend where there is room, so you do not have to look back and forth.
- Light grid, no box round the chart, grey axis text, so the data is the darkest thing on the page.
- Bucket edges are uneven, small near the promised date and wider further out, because almost everything happens in the first week. Equal-width buckets would either hide that or leave tiny bars at the ends.

5. Limitations

- These are patterns, not proof of cause. The distance check is one check, not a model.
- Everything here only covers orders that arrived. The ones that never did are kept separate and never mixed in.
- Orders more than 40 days late are *less* negative, and the error bars do not overlap the ones before them, so it is not just a small-numbers wobble. Those orders have impossible transit times (about 124 days on

average), which makes me think the delivery date was filled in later rather than being real. I could not prove that, so the chart says the cause is not established instead of me making one up.

- Reviews only come from customers who answer the survey, and Figure 4 shows the survey does not go out at the same moment for both groups.

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6. Conclusions

1. The promised date is what matters, not the wait. Before it everything is flat, after it everything falls apart.
2. Delivering early does not buy goodwill. Sixty days of it is worth about two points.
3. The cheapest fix is not faster delivery, it is a more honest promised date. The estimate is set by the platform, so it is the one thing here you can change without touching logistics.
4. The real problem is bigger than these charts show. Orders that never arrive are 2.9% of reviewed orders and 15.3% of all bad reviews, and no delivery-time analysis can see them.

Figures and tables made by the scripts in this folder. Data: Olist Brazilian e-commerce, 2016 to 2018.